A conservative p-value for case studies

A model of a rival argument, transparent assumptions, bounded doubt

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Why would a pro-poor party punish the poor?

In 2013 Uruguay's *Frente Amplio* — a left-wing government built on pro-poor politics — suddenly began cutting cash transfers from families who missed school or clinic visits.

Electoral pressure? Or technocratic policy learning? Rossel et al. collect 8 observations. Seven point one way.

Is 7-of-8 a pattern, or could the rival explain it?

Rossel, Antía & Manzi, *Social Policy & Administration*, 2023.

Two explanations. The same policy shift. Which drove it?

Working theory H_{T}

Electoral competition.

Middle-class voters resent “undeserving” beneficiaries; the FA shifts to hold those voters.

This is the claim Rossel et al. defend.

Rival theory H_{R}

Technocratic policy learning.

Officials learn that enforcement improves education and health outcomes.

A serious alternative — consistent with the literature.

Both stories predict the same policy shift. Only the within-case evidence can break the tie.

The evidence: 7 of 8 observations point to electoral pressure

#	Observation	Supports
1	Opposition campaigned against lax CCT enforcement	both
2	Polls: FA losing affluent voters; redistribution support falling	H_T
3	Op-eds from non-FA pundits pushed enforcement	H_T
4	FA officials cited "public opinion pressure" as reason to shift	H_T
5	Pres. Mujica publicly cited "middle-class pressure"	H_T
6	FA officials unanimously defended lax approach *beforehand*	H_T
7	Government publicized the shift rather than quiet it	H_T
8	Briefings said enforcement was *not* needed for outcomes	H_T

7-of-8 is unlikely given the rival argument, and only implausible counter-example bias would let that argument accommodate the record

	Counter-example bias needed to reach	
p-value upper bound	$p = .05$	$p = .10$
0.001	4.22	6.29

Even if working-theory evidence were four times easier to observe than rival-theory evidence, Rossel et al. would still have strong evidence against the rival ($p \leq .05$).

What our p-value asks

Under the model of the rival argument, how often would we see a record this tilted toward H_{T}?

(Fisher's exact combinatorial probability, under that model.)

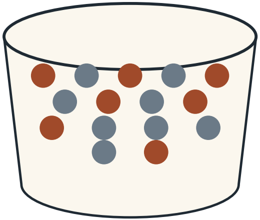
Small $p \rightarrow$ the rival struggles to explain what we see.

Large $p \rightarrow$ the rival is consistent with what we see.

"How often" is a probability.

Where does the probability come from?

Our choice: an urn as a probability model for the rival argument



We posit a probability model — a formal version of the rival's argument; the p-value tells us how poorly that argument fits what we saw.

Not randomization. Not sampling. Not a prior. Akin to a Bayesian's choice of likelihood.

A formal benchmark, finite-sample and combinatorial — one we can state, stress-test, and argue about precisely.

- $m_{\mathrm{T}} = 7$ — observations we actually see, produced by a world where H_{T} is the best explanation
- $m_{\mathrm{R}} = 8$ — observations we do not see, produced by a world where H_{R} is the best explanation

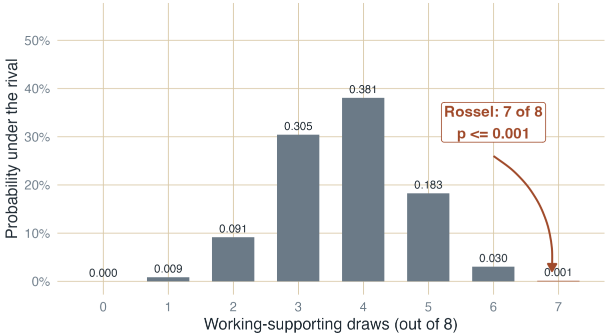
Draw $n = 8$ without replacement — the count Rossel et al. report. Count how many favor H_{T}.

Why $m_{\mathrm{R}} = 8$? We answer in two slides.

Scope: observations treated as independent. ω varies marginal odds; modeling dependence is the natural next step.

Most of the time the rival does *not* produce what we saw

We do not claim the rival is the best explanation. Formalizing a colleague's rival claim — and asking what it would produce — turns disagreement into shared inquiry.



$p \leq 0.001$ for Rossel et al. Under the urn we posited, 7-of-8 is rare.

Our choice: the $+1$ urn — the most conservative benchmark

You just saw what the urn produces. But why $m_{\mathsf{R}} = 8$? We pick:

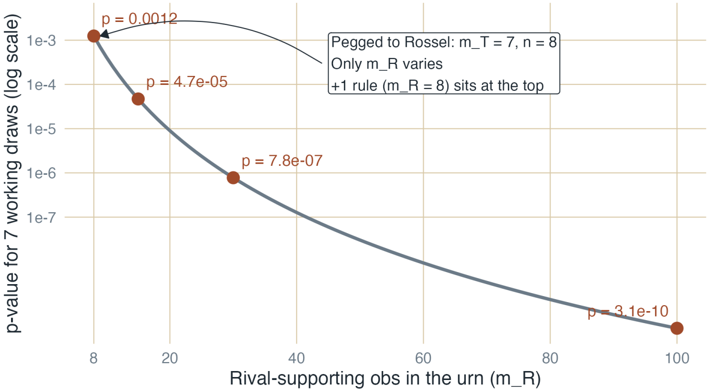
$$m_{\mathsf{R}} \;=\; m_{\mathsf{T}} + 1.$$

Give the rival the most benefit of the doubt the data allow.

Why not go further? A larger m_{R} tilts further toward the rival — but the observed pattern becomes so extreme under that urn that p *shrinks*. $+1$ sits at the ceiling.

Theorem (paper). For any m_{T} and any n, $m_{\mathsf{R}} = m_{\mathsf{T}} + 1$ maximizes p — the same rule whether you have 2 observations or 200.

Larger rival urns shrink p — so $+1$ is the most conservative choice



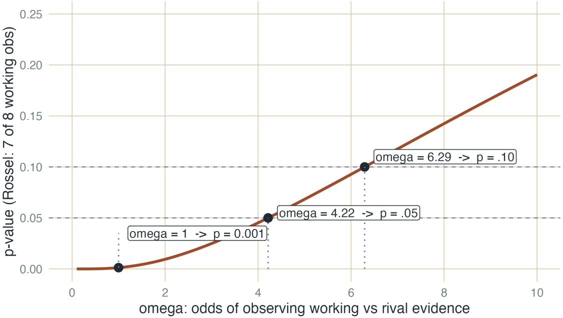
Why $+1$ is conservative: it sits at the top of this curve. (log scale)

How much counter-example bias would it take to overturn the result?

Rival-supporting observations may be harder to find in the record than working-theory-supporting ones. Let ω be the odds ratio.

$\omega = 1$: no counter-example bias.
$\omega > 1$: counter-examples under-represented in what we observe.

For Rossel:
$\omega \geq 4.22$ needed to push p to $.05$.
$\omega \geq 6.29$ needed to push p to $.10$.

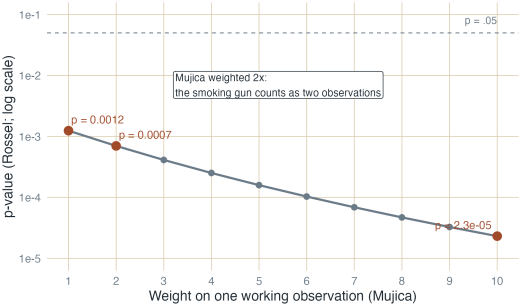


Working evidence would have to be more than $4\times$ easier to observe than rival evidence before the strong argument becomes just barely weak.

One strong observation can count as many

Observation 5 — President Mujica publicly citing middle-class pressure — is a smoking gun. A rival world should not produce it.

We can give it more evidentiary weight: the smoking gun counts as w ordinary observations. The $+1$ rule reapplies to the weighted total, so the rival urn grows with w.



Equal weight $(w = 1)$: $p = 0.001$.
Mujica statement counts double $(w = 2)$: $p = 0.0007$.
Mujica statement counts 10 times $(w = 10)$: $p \approx 2 \cdot 10^{-5}$.

“You are quantifying the unquantifiable.”

A skeptic’s objections

Meaning is interpretive, not a count.

The record leaves out counter-examples we never collected.

Our response

We do not *replace* interpretation — we *summarize* it. Rossel et al interpreted each observation before we counted anything.

We do not *dismiss* the worry — we *bound* it. ω tells us how much the record would have to under-represent counter-examples before p crosses $.05$: more than $4\times$.

The p-value makes the assumptions explicit; the skeptic’s worry, bounded.

The $+1$ urn p-value equips researchers to (1) summarize how much evidence their observations provide against the rival; and (2) reason with colleagues and critics about the consequences of specific worries — missed counter-examples, uneven evidentiary weight.

With thanks to the authors of our running example:

Cecilia Rossel

Florencia Antía

Pilar Manzi

R package: `DrBristol` · `https://bowers-illinois-edu.github.io/DrBristol/`

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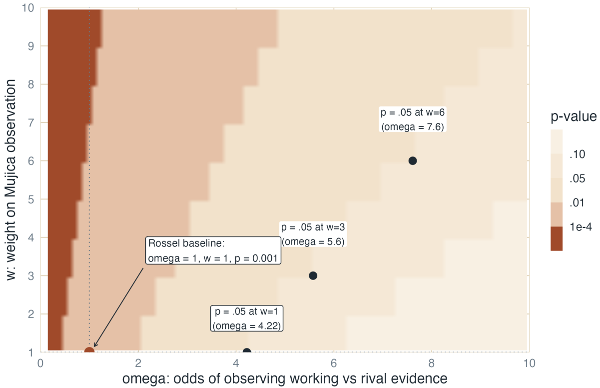
Appendix: joint sensitivity — moving both levers at once



Slides 11–12 varied ω and w one at a time. This plot varies them together.

Darker = smaller p. The three black dots trace the $p = .05$ contour: as w grows, the ω needed to escape rejection grows with it.

About two thirds of this plane still gives $p \leq .05$. Even with Mujica upweighted $6\times$, the rival needs $\omega \approx 7.6$ — nearly an order of magnitude of counter-example bias — to avoid rejection.

For the skeptic who asks:
"what if both assumptions are wrong at once?"

Appendix: the $+1$ theorem (paper, Online Supp.)

Setup. Urn with m_{T} working-supporting balls and $m_{\mathsf{R}} = m_{\mathsf{T}} + c$ rival-supporting balls ($c \geq 1$). Draw n pieces of evidence without replacement; let X count the working-supporting draws. Hypergeometric pmf:

$$P(X = j) \;=\; \frac{\binom{m_{\mathsf{T}}}{j}\binom{m_{\mathsf{T}}+c}{n-j}}{\binom{2m_{\mathsf{T}}+c}{n}}.$$

Theorem 1. For fixed nonnegative integers x, n, m_{T} with $x \leq n$, the tail probability

$$P(X \geq x) \;=\; \sum_{j=x}^{n} \frac{\binom{m_{\mathsf{T}}}{j}\binom{m_{\mathsf{T}}+c}{n-j}}{\binom{2m_{\mathsf{T}}+c}{n}}$$

is maximized over integer $c \geq 1$ at $c = 1$; i.e., at $m_{\mathsf{R}} = m_{\mathsf{T}} + 1$. (The domain $c \geq 1$ encodes "rival has at least one more than we observed." Designs with $c = 0$ sit outside this domain — see Bristol slide.)

Proof sketch: as c grows, the denominator $\binom{2m_{\mathsf{T}}+c}{n}$ grows faster than the numerator $\binom{m_{\mathsf{T}}+c}{n-j}$, so every term in the sum shrinks. Full proof: online supplement.

Appendix: apply $+1$ to Bristol — same verdict, different arithmetic

Suppose we lost Fisher's randomized design. All we know is what Bristol *did*: she identified 4 of 4 milk-first cups.

Fisher's randomization

Design known.

$m_{\mathsf{T}} = 4,\ m_{\mathsf{R}} = 4,\ n = 4$ — fixed by the design.

$\omega = 1$ — guaranteed by the design.

$p = 1/70 = 0.014$.

$+1$ **urn**

Treat it as observational.

$m_{\mathsf{T}} = 4,\ m_{\mathsf{R}} = m_{\mathsf{T}} + 1 = 5,\ n = 4$ — m_{R} chosen by the $+1$ rule.

$\omega = 1$ — assumed.

$p = 1/126 = 0.008$.

Same substantive verdict. The $+1$ rule builds an urn when design cannot.

Appendix: randomization fixes the urn; sensitivity defends our choice

Lady tasting tea (Fisher 1935)

Randomized design.

Bristol identified 4 of 4 milk-first cups.

$m_{\mathsf{T}} = 4,\ m_{\mathsf{R}} = 4,\ n = 4$ — fixed by the design.

$\omega = 1$ — guaranteed by the design.

$p = 1/70 = 0.014$.

No sensitivity analysis needed.

Rossel et al. (2023)

Observational case.

Rossel at al. observed 7 of 8 supporting H_{T}.

$m_{\mathsf{T}} = 7,\ m_{\mathsf{R}} = m_{\mathsf{T}} + 1 = 8,\ n = 8$ — m_{R} assumed by the $+1$ rule.

$\omega = 1$ — assumed, then stress-tested.

$p \leq 0.001$.

Sensitivity analysis does the work randomization would have done.

Randomization chooses the urn for us. Without it, we choose — and defend the choice.

Appendix: when p-value, when Bayesian?

Bayes factors
Fairfield & Charman 2017, 2022

Inputs: likelihood ratios per observation; priors often flat or varied.

Output: weight of evidence (log Bayes factor).

Question: how strongly does this evidence favor H_T over H_R?

Model-based Bayesian
Humphreys & Jacobs 2015, 2023

Inputs: causal DAG; Dirichlet priors over causal-type shares.

Output: posterior over model parameters.

Question: what does the updated model say about my query?

Our $+1$ urn p-value
Lopez & Bowers

Inputs: researcher's coding of each observation as supporting the working theory or rival. (plus maybe w and/or ω)

Output: a conservative p-value. ($p_{+1\ \mathsf{urn}} \leq p$)

Question: how often would the model of observation that gives the benefit of the doubt to the rival produce a record this tilted toward H_T?

Different questions, not competing answers.